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**THE UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads
to the Interstate Transmission System

Docket No. RM26-4-000

COMMENTS OF SMA

The SMA Group welcomes the opportunity to comment on the Federal Energy Regulatory Commission's (FERC) Advance Notice of Proposed Rulemaking (ANOPR) regarding the large-load interconnections.¹ By initiating this timely ANOPR, FERC is taking important steps to improve the efficiency and prioritization of the interconnection process for large loads like AI data centers. SMA believes that without significant reforms, the existing system will be unable to support the rapid deployment of AI data centers necessary for the U.S. to win the global race in AI and digital infrastructure, while ensuring that the regular rate payers have access to affordable and reliable power.

The advanced, high-performing, grid-forming inverters produced by SMA alongside battery energy storage systems (BESS) can play a central role in fulfilling the objectives of the ANOPR. SMA strongly recommends and requests that, as the Commission develops a Notice of Proposed Rulemaking and a final rule, it ensures that grid operators and other stakeholders can fully utilize this essential, adaptable, and high-performing resource. Advanced grid-forming inverters with BESS can reduce the need for costly infrastructure expenditures, which would otherwise be borne by consumers. The capabilities of these systems are versatile; they provide system stability to the grid including provision of inertia, short circuit level, system restoration/black start capability, fast frequency reserves and grid resilience through power quality assurance – basically, everything a rotating turbine provides.

¹ Department of Energy, Letter to the Federal Energy Regulatory Commission (Oct. 23, 2025), <https://www.energy.gov/sites/default/files/2025-10/403%20Large%20Loads%20Letter.pdf>. (DOE Letter).

About SMA

SMA has been actively engaged in grid interconnection processes in the U.S. and globally for decades, collaborating with utilities, regulators, and industry stakeholders to develop and implement efficient, reliable, and standardized interconnection practices. Advanced inverters play a key role in managing the interaction between the power grid and large electrical loads through power electronics. SMA's portfolio includes grid-forming inverters and control technology for BESS that enhance power system stability, provide pro-active voltage and frequency support at critical grid events, as well as enable rapid response to fluctuating loads. SMA's technologies are deployed across a wide range of applications, including utility-scale energy projects and critical infrastructure.² The first ever stability plant, called Blackhillock, is located in the United Kingdom. It delivers 370 MWs of inertia and 116MVA of Short Circuit level – like a gas turbine plant – and utilizes SMA's grid-forming inverters.

SMA has an installed inverter capacity of over 160 GW across more than 200 countries, including more than 45 GW in the U.S. alone. Headquartered in Germany with a branch office since 2008 in Rocklin, California, SMA employs 3,900 individuals in 19 countries, stimulating local economies through high-paying manufacturing jobs and collaborations with regional suppliers and service providers. SMA has an integration partner for the assembly of medium voltage transformers and switch gears in Little Rock, Arkansas for the SMA Medium Voltage Power Stations. SMA's large-scale string-inverter line has a manufacturing partner in Nashville, Tennessee, where PEAK3 is assembled. SMA is committed to expanding U.S. production and supply chains to support critical infrastructure needs.

Comments on ANOPR Principles

The ANOPR sets forth 14 principles to serve as the basis for FERC's rulemaking on the interconnection of large loads to the grid. SMA offers the following recommendations and comments regarding these principles.

Principle	Description – Summary	Recommendations/Comments
1	FERC "jurisdiction should be limited to interconnections directly to transmission facilities, consistent with the Commission's seven-factor test".	No comment.
2	Proposed "reforms should only apply to new loads greater than 20 MW and, for hybrid facilities, where the load is greater than 20 MW".	The performance threshold should be defined to include advanced grid-forming inverters combined with BESS, which may need to be aggregated to exceed 20 MW (in line with the modular build-up

² SMA Grid Forming Solutions. <https://www.sma.de/en/large-scale/grid-forming-solutions>.

		of data center racks). This technology supports faster project implementation and enhances power system stability. SMA considers the connection capacity to be the relevant metric for evaluation and not the capacity of one single inverter.
3	"[L]oad and hybrid facilities should be studied together with generating facilities".	SMA strongly supports the simulation of load and hybrid facilities together as we find large gaps today in addressing the risks on both the grid and AI data center (load) sides, in that the load-side ignores real risks to the grid and vice versa.
4	"[L]ike generating facilities, load and hybrid facilities should be subject to standardized study deposits, readiness requirements, and withdrawal penalties".	SMA recommends that FERC consider differentiated and proportionate study deposits and readiness requirements for grid-supportive technology providers. Advanced grid-forming inverter and BESS solutions contribute directly to system stability and can reduce the need for costly network reinforcements. To ensure market access for such technologies and to encourage their deployment in support of broader grid reliability goals, SMA suggests reducing initial deposit requirements by approximately 15 percent for applicants with a proven track record of delivering grid-supportive performance. This approach would lower barriers to entry while maintaining effective safeguards against speculative projects.
5	"[H]ybrid facilities should be studied based on the amount of injection and/or withdrawal rights requested...[to] provide[] incentives for co-location with new generation facilities and ensure[] efficient buildout of the transmission system".	A key consideration for hybrid facilities serving large, variable loads such as data centers is the highly dynamic load profile, which can pose challenges for maintaining system stability. BESS with dynamic dispatchability can complement existing generation by providing fast, responsive flexibility to manage short-term fluctuations, support frequency stability, and optimize grid operations.
6	"[H]ybrid interconnection shall be required to install the system protection facilities necessary to prevent unauthorized injections or withdrawals that exceed the respective rights".	SMA's perspective is that penalties are not necessary if system protection settings are well coordinated and operate effectively at the overall system level. Disconnection itself is not beneficial for the customer. There are existing requirements on dynamic active power limitation to prevent grid overloading which may serve as a starting point.

7	The “interconnection study of large loads that agree to be curtailable and hybrid facilities that agree to be curtailable and dispatchable should be expedited”.	Controllability (in particular curtailability and dispatchability) with SMA technology is already field-proven and state-of-the-art (system operators can manage it) using, for example, the IEEE 2030.5 (SEP2) protocol. Preferably, controllability should be assessed at the interconnection point rather than at individual components, allowing owners maximum flexibility in implementation.
8	“[L]oad and hybrid facilities should be responsible for 100% of the network upgrades that they are assigned through the interconnection studies”.	Incentives should aim to minimize the impact on public infrastructure. Co-location can help “minimize the need for costly network upgrades,” and BESS with grid services can further reduce these costs. The modular power electronics approach simultaneously supports faster interconnection of large loads and power system stability.
9	“[T]o the extent the interconnection customer is not the transmission owner, the interconnection customer shall be afforded the same (or equivalent) option to build as currently provided to generator interconnection customers”.	No comment.
10	“[A]n existing generating facility that seeks to enter a partial suspension to serve a new load at the same location must go through a system support resource (SSR)/reliability must run (RMR) type study”.	No comment.
11	“[U]tilities serving large loads, including those at hybrid facilities, should be responsible for transmission service based on their withdrawal rights, as that value amount reflects the quantity of capacity and energy that is being transmitted across the transmission system to the load”.	Active power from local facilities could temporarily support the load if appropriately compensated, for example through peak shaving. Consideration should be given to measures that ensure capabilities are available during system constraints.
12	“[U]tilities serving large loads, including those at hybrid facilities, should be responsible for ancillary services based on peak demand, without consideration of any co-located generation”.	Responsibility should remain with the utility, but care must be taken to avoid slowing down interconnection of critical load projects if this becomes a bottleneck. SMA suggests allowing a “priority setup” to be connected, where facilities



		with their own infrastructure that already supports grid stability - such as BESS with advanced grid-forming capabilities - can provide all relevant ancillary services and receive appropriate incentives (e.g. compensation fees or easy connection process).
13	Promulgation of “a plan to implement” the ANOPR’s proposed changes.	SMA supports prioritizing large loads, especially when co-located with generation, provided that power system stability is maintained and further price increases are avoided. Large, highly dynamic loads pose stability risks, and additional grid expansion is costly. Therefore, grid-stabilizing rather than grid-following technologies should be used to avoid unnecessary network upgrades and to ensure secure system operation. Cost responsibility should be ring-fenced so that general ratepayers are not burdened. SMA is prepared to contribute both technical expertise and proven technology to support this transition.
14	“[U]tilities serving large loads must meet all applicable [North American Electric Reliability Corporation] reliability standards and [Open Access Transmission Tariff] provisions”.	High performance grid-forming inverters are fully aligned with these requirements. For example, all of SMA’s BESS solutions routinely meet applicable NERC reliability standards and OATT provisions, providing reliable support for consultants and system operators.

ANOPR and Consumer Affordability

SMA supports the ANOPR objective to ensure that U.S. consumers and industrial customers, including strategically important loads such as data centers “have access to affordable, reliable, and secure energy.”³ The ANOPR, including by incorporating SMA’s recommendations, can significantly advance this goal:

- The ANOPR’s Seventh Principle, by enabling large loads to curtail their demand from the grid provides the opportunity to reduce costs by accelerating interconnections and avoiding expensive infrastructure investments.⁴

³ DOE Letter.

⁴ Walsh, S., M. Farmer, A. Zevin, K. Smaczniak, G. Daly, N. Lobel, A. Clements, Y. Guo, and T. Profeta, How DOE’s Proposed Large Load Interconnection Process Could Unlock the Benefits of Load Flexibility. NI PB 25-03. Durham, NC:

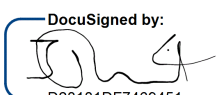
- The Eighth Principle, by potentially encouraging co-location, including by deploying BESS, can reduce the need for costly grid upgrades.
- In developing a plan for implementing the ANOPR's proposed reforms, pursuant to the Thirteenth Principle, there is an opportunity to ensure that ratepayers are not burdened with the costs of interconnection and co-location of large loads.

Conclusion

SMA appreciates the Department of Energy and the Commission's leadership in addressing the challenges of large load interconnection. SMA strongly supports FERC's continued efforts to develop a rulemaking that incorporates the principles outlined in the ANOPR, while ensuring that advanced technologies like grid-forming inverters and BESS are recognized for their role in supporting process expedition while ensuring a reliable and resilient power system. SMA looks forward to continued engagement with the Commission and stakeholders on this critical issue.

Respectfully submitted,

Kassel, November 19th, 2025

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